

#hack

#behavioral-geometry

#ukb-pentadic-calculus

#dyad

Karl Marx commits a McGilchrist sin of compression into dyads: "Our epoch, the epoch of the bourgeoisie, possesses, however, this distinct feature: it has simplified class antagonisms. Society as a whole is more and more splitting up into two great hostile camps, into two great classes directly facing each other – Bourgeoisie and Proletariat."

Excerpt From

The Communist Manifesto - with full original text by Karl Marx

Rupert Matthews

<https://books.apple.com/book/id498418987>

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#triad

Einstein reduced the cosmos to a triad. His fault is being classically oriented. Our more delicate sensibilities admit time & work to make a pentadic stack, aligning perfectly with our calculus: $e, c, m, t[x, y, z], w$. So Einstein doesn't distinguish geometry from behavior.

But since we are generalizing beyond relativity, we need protagonists {raindrops or tumbleweeds, not just photons at the $\max(\text{signal speed})$ } to traverse a massive combinatorial search space (much more complex terrain than the cosmic manifold); but otherwise, we conserve energy & mass by the relativistic constraints he proposed.

He provided the hard boundaries that clarify Demis Hassabis "substrate" for AI: massive combinatorial search space (not in Einstein's equation, but in his manifolds of space-time: and hence geometry-behavior), unambiguous optimization ($\max[\text{mass} \times \text{signal} = \text{speed/energy}]$), and conservation laws from $e=mc^2$ and thermodynamics (data is energy is mass is signal is some transformation of these elements)

This invites simulation, parametrization, matrices, weights, tensors, TPUs, GPUs, and "compute". So we can comfortably move amongst Heisenberg, Prigogine, Vogelstein, Dostoevsky, and Nietzsche without shattering.

We're finally emancipated from relativity, quantum mechanics, and the grid-lock betwixt :)

#pentad

Our substrate & resource is e. Latency & responsiveness is c. Reliability is m. And we generalize from relativity, to thermodynamics, and mechanics to yield w, the integral and ledger that conserves all quantities.

It doesn't confuse the map (epistemology/z) with the territory (ontology/epsilon)

- (x, y, z, ϵ)
- $y(t \mid x) + \epsilon$
- $\frac{dy_x}{dt}$
- $\frac{dy_{\{\bar{x}\}}}{dt} \pm z \sqrt{\frac{d^2 y_x}{dt^2}}$
- $\int y_x dt + \epsilon_x t + C_x$

We declare that individual (x) and group ($\bar{x}$) behavioral dynamics can be fully derived from geometry, by flipping axes of reference.

If geometry had already captured landscape & geology (z = latitude, x = longitude, y = altitude, t = geological time, ϵ = perturbation or in Karl Marx's language "a spectre to be exorcised"), then our pentadic-calculus provides the dynamics

Local & global maxima & minima, saddle points, escarpments, basins, rifts, contexts, concavity, raindrops, brooks, bifurcation, etc can be expressed with elegant and compressed notation

But when we change time scale from geological time to ms, s, min, h, d, w, mo, y, d, c, etc? Then we are describing those who are preponderant under the atmospheric cosmogony of the classical universe; those who are in a good-soul, bright & cannot find any symptoms whatsoever to symbolize the verbosity!

Handel might have endured perturbation under Italian opera singers. But when circumstances in London forbade secular music, and when he suffered threats to his livelihood, something in him transformed the Italian secular genre into a good 'ol' English oratorio. And the rest is history. For precision, we can apply some behavioral geometry to capture that phase in the composers life.

And when we gain confidence in our approach, we may find ourselves describing Bach as a global minima, Ludwig as gradient ascent, and Mozart & Shakespeare as saddle points !!

McGilchrist's two-part thesis then is upgraded from "The brain" and "The World" to "Raindrops" and "Droptop". Part 1 described asymmetry of the brain, left vs right hemisphere, language -> music -> semantics, two words, primacy of right, and usurpation by left. Part 2 postulates mimesis as a mechanism, and history of

ancient, baroque-reformation/renaissance, classical/enlightenment, romantic, and modern/post-modern as consequences.

Ours is a corrective. Raindrops fall onto a landscape and we have orientation invariance (symmetry) between right-sided vs left sided raindrops (think of an arbitrary dividing line in landscape). That's part 1. And the mechanism in part 2 is stochastic gradient descent. Our pentadic calculus populates chapters 2-6 of part 1. But we keep chapters 8-12 with the same names, but with a complete override of narrative: its Prigogine dissipation all the way down into modern times.

After all, it's the costly PFC under strict constraints of the 20W human brain that have demanded such dissipation, initially sparked by a serpentine perturbation in the garden of Eden (a local minima). Think of it as ***Das Kapital***. We literally outsourced our souls to the devil; and like soul-less zombies (THE mode of alienation), we rove about the earth maximizing capital gains at any cost to our fellow man, ecosystem, cultures, individuality, etc. That was the primal faustian bargain all in exchange for what? A little money? The Baganda did it (**Tetwaagala tunyonyola; twaagala ki? Twaagala ssente!**). The Dutch did it. But the English remind us that power > money (New Amsterdam -> New York, and Anglo-Boer war).

Karl Marx, then, wishes to return humanity to Eden. Our calculus recognizes the frailty inherent in that craving. It's nostalgia with all its errors. Surely, what did he gain from his doctoral research into Epicurus (soul & destiny) vs. Democritus (raindrop-droptop)?

Zarathustra, by contrast, embraces an eternal recurrence of down-going & up-going, with singing, dancing, and exuberance !! What a difference from medieval Dante!!!! It's this Nietzschean credo that would suffocate *Das Kapital*, humanity would renew its soul (**akomyawo ememe yange, oyo Dionysus** — my personalized god!). Move-over KPIs, moveover Wallstreet, fuck you guys, I'm going home.. gradient descent, and ascent, and descent, eternally recurrently!

Shouldn't we be invited, then, to chart out the entire landscape, map it all out in its entirety? Embrace the Dionysian epsilon, crank-up the z-score in the spirit of a **perspectivism & pessimism beyond good & evil**, that ensures we are never trapped in the comforting Psalm 23 local minima?

That way, when we outsource the PFCs chores to AI (unstoppable gradient descent in light of the adversarial dynamics amongst adversaries, reminiscent of the nuclear arms race), we know the landscape on which it will optimize the SGD. Ukubona!!!

—

"The history of all hitherto existing society is the history of class struggles. Freeman and slave, patrician and plebeian, lord and serf, guild-master and journeyman, in a word, oppressor and oppressed, stood in constant opposition to one another, carried on an uninterrupted, now hidden, now open fight, a fight that each time ended, either in a revolutionary reconstitution of society at large, or in the common ruin of the contending classes.
"

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It's taken me about 20 years **Ukubona** the flaw in this brilliant rhetoric. Can a raindrop be accused of being part of a class war when all it does is abide the local conditions? I guess, then, over large swaths of time earlier raindrops curved the landscape and created its features.

No need to express any rage about that! Just abide the terrain today, abide SGD, never-ever accept a local minima (ie never feel obligated to a Master, never-ever consider yourself an emissary).

You only have the laws of physics to abide: stochastic gradient descent. Even your PFC has the same laws to abide by.

And that all, folks!